Tutorial 9

Arrays

Overview

- Revision:
    - Strings

- 1D arrays
    - swap/reverse
    - isPalindrome

Revision: string

- In C++ a string is an array of characters
- A **small** string is stored on the **stack**, while all other strings are stored on the **heap**.
- Access the first character of a string that is defined by:
  - string myString = "HelloWorld";
  - myString[0] access the first character in the string,
  - &myString is the reference to where the first character of the string is allocated in memory.
- Pointers to Strings
  - string myString = ”Hello World!”;
    - Stored on either the heap or stack: **&myString**
    - First character: **myString[0]**
  - string *strPtr = new string(”Hello, world!”);
    - Stored on the heap: ***strPtr**
    - First character: **(*strPtr)[0]**

Arrays: Introduction

- Definition:
  - A list of homogenous (same datatype) elements
- Syntax
  - **Declaration**: <data_type> <array_name>[array_size]
    - int v[12];
  - **Initialization**: <Declaration> = { V_1, V_2, …, V_{array_size}}
    - int w[] = {1,2,3,4}
    - int w[4] = { 1,2,3,4 };
    - **Empty Members**:
      - <Declaration> = { V_1, V_2, …, $V_{n<array\_size}$}
      - int w[4] = {1,2};
  - **Access**: <array_name>[subscript]
    - The word <u>subscript</u> can sometimes be used <u>interchangeably</u> with <u>index</u>
    - Subscript starts at 0, the subscript of the last element of the array is **array_size-1**
    - W[0] = 12; //access the first element of the array

Arrays: Swap

Problem:

- Given an array of type int write separate functions that perform the following:
  - Swap the values at the given subscripts/indices
  - Swap the positions of the highest 2 values
  - Swap the positions of the highest and lowest values
- Update the implementations of the previous functions to use pointer notation instead of square brackets [] .
  - 2 steps to note is incrementing first then dereferencing
  - *(array_name + incremetor)

Arrays: Reverse

Problem:

- Given an array of type int write separate functions that perform the following:
  - Reverse the order of the array
  - Checks if the array isPalindrome
    - A palindrome is a word, number, phrase, or other sequence of symbols that reads the same backwards as forwards,
    - such as **madam** or **racecar**
  - Uses the reverse order function to test isPalindrome

END